

Normative Formant Data for Korean Vowels: A Resource for Clinical and Pedagogical Application

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Received: December 24, 2025

Revised: February 27, 2026

Accepted: March 6, 2026

Objectives: This study aims to establish large-scale normative data for vowel formant frequencies and speech intelligibility (SI) in healthy Korean adults, providing a critical resource for clinical and pedagogical applications in speech-language pathology and voice science.

Methods: A comprehensive cross-sectional analysis of 4,235 healthy Korean adults across three distinct age groups was conducted. We systematically measured multiple acoustic variables including F1, F2, vowel space areas (VSA_3, VSA_4, VSA_5), Vowel Articulation Index, Formant Centralization Ratio, and F2 ratio using customized Praat scripts, and assessed subjective SI through perceptual measures to establish reliable baseline data.

Results: Significant age-related differences were found in all variables except the F2 ratio ($p = .14$). Significant sex-based differences were confirmed for all variables except the F1 of /o/ ($p = .327$). All acoustic metrics correlated significantly with SI, with VSA_5 demonstrating the highest correlation among all vowel space metrics ($r = -.56, p < .01$), suggesting its potential clinical utility as a complementary objective measure for speech assessment.

Conclusion: This study provides robust, large-scale normative data confirming that age and sex significantly influence Korean vowel acoustics, with older age groups showing notable reductions in VSA and SI. Among the acoustic metrics examined, VSA_5 showed the strongest correlation with perceived speech intelligibility, indicating its potential as a meaningful objective complement to subjective SI assessment. These findings may offer a practical reference for clinicians in voice assessment, differential diagnosis, and monitoring rehabilitation progress in individuals with speech and voice disorders.

Keywords: Formant frequencies, Vowel sounds, Speech intelligibility

Formants, particularly the first (F1) and second (F2) formants, are critical acoustic features reflecting vocal tract resonance and tongue position (Heeringa et al., 2009). Establishing standard values for the five common Korean vowels (/a, e, i, o, u/) is essential for both clinical and research applications. These values are typically measured using robust software like Praat, which facilitates large-scale acoustic analysis (Boersma & Weenink, 2023).

In clinical practice, acoustic analyses serve as objective complements to subjective auditory-perceptual assessments of Speech Intelligibility (SI) (Garcia & Dagenais, 1998; Thompson et al., 2023). To evaluate vowel articulation quantitatively, the Vowel Space Area

(VSA) is widely employed. This geometric representation, plotted using F1 and F2 values, provides insights into vowel quality differences across speakers and languages (Neel, 2008; Wieland et al., 2015). VSA was calculated using three configurations: VSA_3 (triangle) based on /a/, /i/, and /u/; VSA_4 (quadrilateral) based on /a/, /e/, /i/, and /u/; and VSA_5 (pentagon) based on all five monophthongs (/a/, /e/, /i/, /o/, /u/). To enhance measurement precision, the Euclidean distance between each vowel pair in the F1-F2 acoustic space was computed, and the area of each configuration was computed from the inter-vowel distances among the respective coordinate vertices (Seong, 2005). While VSA characteristics can vary by

language structure and age (Bradlow, 1995), sensitive indices such as the Vowel Articulation Index (VAI) and Formant Centralization Ratio (FCR) have been introduced to enhance reliability, especially in pathological contexts (Liu et al., 2021; Sapir et al., 2009).

Previous studies have explored the relationship between these acoustic measures and communicative function. Research has demonstrated moderate to strong correlations between VSA metrics and subjective SI in both neurotypical individuals and those with dysarthria, suggesting that acoustic variables are valid predictors of intelligibility (Weismer et al., 2001). In Korea, several studies have investigated the acoustic characteristics of Korean vowels and their clinical relevance. A foundational acoustic analysis of Korean monophthongs characterizing the F1 and F2 structure of the Korean vowel space provided essential normative reference data (Seong, 2004). Sex-based differences in vowel formant frequencies have also been reported using a natural speech corpus, providing normative acoustic data for Korean adults (Yoon & Kim, 2015). Furthermore, VSA was found to be significantly reduced in sentence-level production in children with articulation disorders, with no significant correlation observed between VSA and speech intelligibility, suggesting that intelligibility is influenced by multiple acoustic factors beyond vowel space alone (Eom & Shin, 2018). In speakers with spastic dysarthria, vowel space metrics and formant transition parameters were significantly reduced compared to healthy controls, with high correlations observed between the two measures ($r > .8$), supporting the validity of acoustic vowel space analysis as an objective tool for assessing speech intelligibility (Kim et al., 2014; Nam & Park, 2021). Despite these contributions, large-scale normative data integrating multiple VSA configurations and SI measures across age groups and sex remain limited in Korean.

Therefore, this study primarily aimed to establish normative formant frequencies for Korean vowels in a healthy demographic. We specifically investigated: 1) differences in vowel formants, VSA variables, and SI across distinct age groups; 2) potential variations based on sex; and 3) the correlation between VSA variables and SI.

METHODS

Participants

This study was conducted by recording sustained vowels and

Table 1. Demographics of participants included in present study

Group	G2030	G4050	G6070
Total (N)	1,006	1,940	1,289
Female	675 (67.1%)	1,294 (66.7%)	812 (63.0%)
Male	331 (32.9%)	646 (33.3%)	477 (37.0%)
Age (Mean $\pm$ SD)	29.8 $\pm$ 6.9	50.6 $\pm$ 5.9	66.6 $\pm$ 5.6

Values are presented as number (percentage).

connected speech from patients who visited the Department of Otolaryngology at Pusan National University Hospital between January 2019 and December 2023. This cross-sectional study analyzed 4,235 native Korean speakers (aged 20–81), stratified into three groups: G2030 (20–39 years), G4050 (40–59 years), and G6070 (≥ 60 years). Inclusion criteria required the absence of laryngeal or neurological pathologies (e.g., dysarthria, hearing loss) and a GRBAS (Grade, Roughness, Breathiness, Asthenia, Strain) grade of 0. All criteria were verified through medical records and interviews. Demographic details are presented in Table 1. This study was approved by the Institutional Review Board (IRB) of Pusan National University Hospital and adhered to the established research ethics and standards (IRB No. 2312-005-133).

Voice Recording Procedure

Speech samples were recorded in a soundproof booth using CSL 4500 and analyzed via Praat (v6.3.17) for formant analysis. All participants received standardized instruction from a certified speech-language pathologist (SLP) prior to recording, who provided direct modeling to ensure that each participant produced the target vowels with a neutral pitch and comfortable intensity level, thereby minimizing acoustic variability attributable to dialectal characteristics. Participants articulated five isolated vowels (/a, e, i, o, u/) and read the “Walking” passage three times each. Each vowel production was recorded for formant analysis, and the “Walking” passage was recorded for speech intelligibility assessment. For the acoustic analysis, VSA, VAI, and FCR were measured using the five vowels. Isolated vowels were analyzed to minimize coarticulation effects and segmentation errors associated with extracting vowels from continuous speech (Skoruppa et al., 2015). Voice samples were recorded and saved in ‘wav’ format with a sampling rate of 44,100 Hz.

Formant Analysis & Reliability

Formant extraction and VSA variable calculations were performed using custom Praat scripts (Derdemezis et al., 2016). Following the Praat manual recommendations to account for vocal tract size differences, the maximum formant frequency was set to 5,000 Hz for male and 5,500 Hz for female voices (Appendix 1). Due to script configuration constraints, analyses were conducted separately by sex and each of the five vowels (Boersma & Weenink, 2023). Each of the five vowels was sustained for 2 seconds (2,000 ms) during recording. For formant analysis, a stable 500 ms segment was extracted from each vowel and saved as an individual file. Praat's selection function allows for precise extraction of an exact 500 ms segment. Files with measurement errors were excluded, accounting for less than 0.5% of the total data. Intra-rater reliability, assessed on 20% of the dataset after two weeks, demonstrated high consistency ($r = .98-.99$) with no significant differences.

SI Rating

Speech Intelligibility (SI) was assessed by three experienced SLPs (>7 years clinical experience) using a 5-point scale derived from the STAND (Screening Test for Aphasia and Neurologic-Communication Disorders) instrument (Kim et al., 2009), ranging from 1 (normal) to 5 (extreme). Ratings were based on both vowel and paragraph reading tasks. Assessments were conducted across four sessions in a noise-controlled room to prevent judgment errors arising from exhaustion. To verify reliability, 400 randomly select-

Table 2. Intra- and inter-rater reliability for speech intelligibility assessment

	Rater 1	Rater 2	Rater 3
Rater 1	.827**	.788**	.899**
Rater 2		.812**	.854**
Rater 3			.901**

Values on the diagonal (in parentheses) represent intra-rater reliability. Off-diagonal values represent inter-rater reliability.

** $p < .01$.

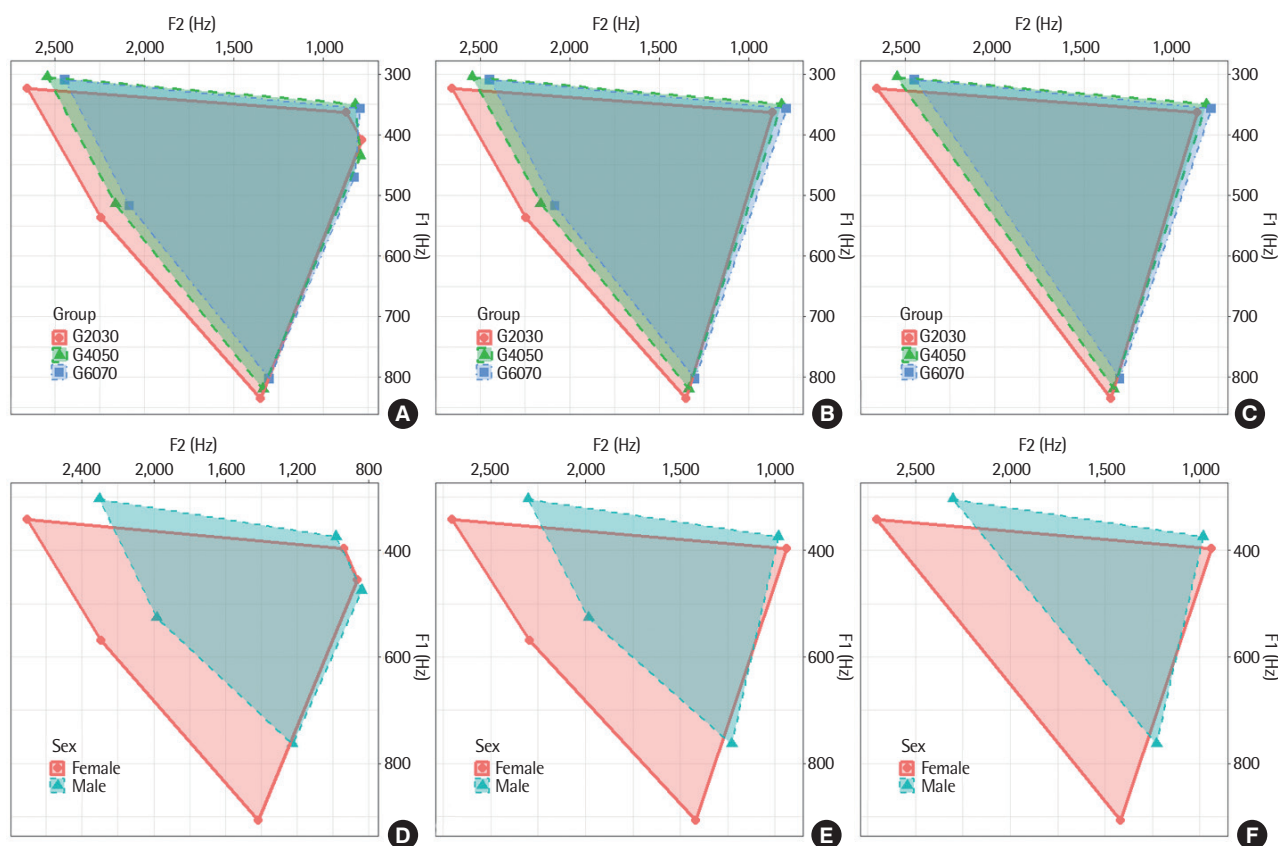


Figure 1. Visualization of VSA variables based on age and sex groups.

(A) Pentagon-shaped vowel space by age group, (B) Quadrilateral-shaped vowel space by age group, (C) Triangle-shaped vowel space by age group, (D) Pentagon-shaped vowel space by sex, (E) Quadrilateral-shaped vowel space by sex, (F) Triangle-shaped vowel space by sex.

ed samples (10%) were used for inter-rater reliability, and a subset of 80 samples was re-evaluated after two weeks for intra-rater reliability.

Statistical Analyses

Statistical analyses were performed using R (v4.3.1) and SPSS (v20.0), with statistical significance set at $p < .05$. Intra- and inter-rater reliability were assessed using Intraclass Correlation Coefficients (ICCs). Outliers and extreme values (> 1.5 and > 3 times the interquartile range, respectively) were excluded to prevent distortion. Group differences were analyzed using one-way ANOVA with Bonferroni post-hoc tests (for age) and independent t -tests (for sex). Spearman correlations were employed to examine the relationship between VSA variables and SI.

RESULTS

Inter-rater/intra-rater Reliability

The inter-rater reliability for SI scores indicated moderate to high agreement (mean ICC = .847; range: .788–.889), while intra-

rater reliability demonstrated high consistency (range: .812–.901) (Table 2).

Comparison of Acoustic Characteristics and SI by Age Group

Significant age-related differences ($p < .01$) were observed in formant frequencies (F1, F2), all VSA measures, VAI, FCR, and SI scores (Figure 1, Table 3). In contrast, the F2 ratio showed no significant variation across groups ($p = .14$). Post-hoc Bonferroni comparisons confirmed progressive age-related changes, with the oldest group (Group 3) differing significantly from younger groups in most variables ($p < .001$). F2 decreases were most pronounced in high vowels /i/ and /e/, while all VSA metrics showed significant reductions accompanied by increased VAI and decreased FCR, indicating vowel centralization. As illustrated in Figure 1, the vowel space progressively contracted with advancing age across all three configurations (pentagon, quadrilateral, and triangle), consistently in both female and male speakers. This contraction was reflected in significant age-related changes in VAI and FCR (both $p < .01$), indicating progressive vowel centraliza-

Table 3. Comparison of formant-based acoustic measurements across age groups

Variables	Group			<i>F</i>	<i>p</i>
	G2030	G4050	G6070		
a_F1	834.9 ± 185.0	819.2 ± 163.3	802.5 ± 159.9	10.687	< .01
a_F2	1,349.5 ± 206.7	1,335.7 ± 213.8	1,301.1 ± 213.6	16.847	< .01
e_F1	536.8 ± 115.2	513.9 ± 88.7	517.3 ± 87.3	20.005	< .01
e_F2	2,245.2 ± 265.6	2,162.9 ± 273.1	2,083.3 ± 258.1	104.471	< .01
i_F1	323.9 ± 64.8	304.7 ± 51.0	309.4 ± 50.1	41.668	< .01
i_F2	2,655.6 ± 289.0	2,545.0 ± 276.6	2,446.3 ± 279.4	157.709	< .01
o_F1	408.8 ± 81.5	434.5 ± 82.2	470.2 ± 93.5	150.682	< .01
o_F2	783.5 ± 150.4	790.4 ± 127.4	824.1 ± 126.4	33.766	< .01
u_F1	362.8 ± 69.4	349.8 ± 59.4	357.1 ± 62.3	15.232	< .01
u_F2	868.9 ± 165.6	817.1 ± 147.3	791.2 ± 140.2	77.521	< .01
VSA_3	441,071.3 ± 193,646.2	425,678.8 ± 173,308.6	389,660.3 ± 164,485.0	27.257	< .01
VSA_4	474,312.5 ± 185,574.0	453,187.9 ± 168,074.8	418,292.4 ± 154,961.7	33.162	< .01
VSA_5	506,662.0 ± 195,882.9	482,233.7 ± 176,450.2	439,342.9 ± 159,328.0	44.19	< .01
VAI	1.21 ± 0.13	1.20 ± 0.12	1.18 ± 0.12	16.028	< .01
FCR	0.84 ± 0.10	0.84 ± 0.09	0.85 ± 0.10	12.793	< .01
F2 ratio	3.15 ± 0.59	3.19 ± 0.54	3.17 ± 0.56	1.966	.14
SI	1.79 ± 0.43	2.12 ± 0.44	2.78 ± 0.50	142.681	< .01

Values are presented as Mean ± SD.

VSA=vowel space area; VAI=vowel articulation index; FCR=formant centralization ratio; F2 ratio=ratio of the second formant frequency of the vowel /i/ to the second formant frequency of the vowel /u/; SI=speech intelligibility.

tion in older speakers, while the F2 ratio showed no significant group differences ($p=.14$), suggesting that front-to-back vowel distinction was relatively preserved. SI demonstrated the most pronounced age-related change ($F=142.681, p<.01$), with scores increasing substantially from the youngest ($1.79 \pm .43$) to the oldest group ($2.78 \pm .50$), reflecting a marked deterioration in speech intelligibility with advancing age.

Table 4. Comparison of formant-based acoustic measurements between sex groups

Variables	Sex		<i>t</i>	<i>p</i>
	Male	Female		
a_F1	705.9±151.5	876.4±145.1	-35.74	<.01
a_F2	1,152.7±183.4	1,420.3±163.8	-48.41	<.01
e_F1	479.2±97.7	541.9±87.3	-21.30	<.01
e_F2	1,918.6±217.8	2,283.5±207.9	-53.34	<.01
i_F1	288.6±54.6	322.2±51.4	-19.80	<.01
i_F2	2,264.9±242.4	2,685.7±192.8	-61.58	<.01
o_F1	441.1±88.1	438.3±88.9	.98	.327
o_F2	749.7±156.2	824.8±112.5	-17.97	<.01
u_F1	347.8±70.0	358.9±58.6	-5.45	<.01
u_F2	769.6±175.5	848.7±130.9	-16.54	<.01
VSA_3	282,851.4±127,215.7	489,227.2±156,716.2	-43.30	<.01
VSA_4	315,095.7±122,682.5	516,854.6±148,691.7	-44.43	<.01
VSA_5	337,390.6±129,626.7	546,919.2±156,449.7	-43.81	<.01
VAI	1.17±0.13	1.21±0.11	-10.62	<.01
FCR	0.86±0.11	0.83±0.08	11.02	<.01
F2 ratio	3.05±0.61	3.23±0.52	-9.84	<.01
SI	2.45±0.56	2.13±0.58	16.738	<.01

Values are presented as Mean ± SD.

VSA=vowel space area; VAI=vowel articulation index; FCR=formant centralization ratio; F2 ratio=ratio of the second formant frequency of the vowel /i/ to the second formant frequency of the vowel /u/; SI=speech intelligibility.

Table 5. Correlations among acoustic variables and SI

	Vowel_3	Vowel_4	Vowel_5	VAI	FCR	F2-ratio	SI
Vowel_3	1	.920**	.904**	.490**	-.491**	.386**	-.495**
Vowel_4		1	.978**	.528**	-.518**	.405**	-.540**
Vowel_5			1	.462**	-.453**	.306**	-.562**
VAI				1	-.979**	.747**	-.315**
FCR					1	-.740**	.303**
F2-ratio						1	-.172**
SI							1

VAI=vowel articulation index; FCR=formant centralization ratio; F2 ratio=ratio of the second formant frequency of the vowel /i/ to the second formant frequency of the vowel /u/; SI=speech intelligibility.

** $p<.01$.

Analyzing Formant-based Acoustic Measurements and SI between Sexes

Except for the F1 of /o/, significant sex differences were observed across all variables, including formant frequencies, VSA metrics (VSA_3, VSA_4, VSA_5), VAI, FCR, F2 ratios, and SI scores (Table 4). Male speakers demonstrated markedly smaller vowel spaces than females across all three VSA measures (all $p<.01$), with reductions of approximately 43–44%. This contraction was further supported by significantly lower VAI and higher FCR in males (both $p<.01$), indicating greater vowel centralization, as well as a significantly lower F2 ratio ($p<.01$), reflecting reduced front-to-back vowel distinction. In contrast, SI scores were significantly higher in males ($p<.01$), suggesting greater speech intelligibility impairment compared to females.

Correlations between VSA Variables and SI

As detailed in Table 5, all acoustic variables demonstrated significant correlations with SI ($p<.01$). VSA_5 exhibited the strongest correlation with SI ($r=-.56$), while the F2 ratio showed the weakest ($r=-.17$). The acoustic measures also showed high internal consistency. The three VSA measures were highly inter-correlated (mean $r=.93$; range: .90–.98), and the derived indices (VAI, FCR, F2 ratio) also showed strong, though varied, inter-relationships.

DISCUSSION

This study confirmed significant age-related variations in vowel formants, VSA metrics, and SI scores within a large Korean co-

hort. Notably, the older adult group exhibited substantial reductions in both VSA and SI compared to younger groups. These acoustic changes are likely attributable to physiological factors associated with aging, such as vocal tract lengthening, altered posterior cavity dimensions, and reduced articulatory precision (Eichhorn et al., 2018). Complex interactions among these anatomical and kinematic changes may differentially influence acoustic characteristics across individuals.

Our findings align with a substantial body of literature reporting age-related shifts in formant frequencies (Scukanec et al., 1991; Xue & Hao, 2003). While widespread reductions in F1 and F2 and articulatory centralization are commonly reported (Decoster, 1999; Rastatter & Jacques, 1990), our observation of increased formants for /o/ supports the nuance that acoustic changes may be vowel-specific rather than uniform (Xue et al., 1999). Conversely, some studies maintain that formant frequencies remain relatively stable across the lifespan (Fletcher et al., 2015; Sebastian et al., 2012), highlighting the complexity and variability of aging effects on speech production.

Our finding of an age-related decrease in VSA aligns with previous research demonstrating smaller acoustic vowel spaces in older adults (Kent & Rountrey, 2020). This compression is widely attributed to physiological changes in the vocal tract, declining muscle control, and altered speech-motor coordination. Notably, the lack of significant VSA differences observed between middle-aged and older adult groups in some studies suggests a potential pattern: articulatory performance may decline after a certain age and subsequently plateau, or remain “maintained,” at that reduced level.

Consistent with findings on deteriorating oral function (Ferand, 2001), the observed vowel centralization (reflected in VAI and FCR) became more apparent with age. This trend mirrors findings in pathological aging, such as Parkinson’s disease, where VAI decreases and FCR increases (Skodda et al., 2011). The age-related decline in SI scores further reinforces this, aligning with extensive research in dysarthria and ALS where reduced vowel spaces are consistently correlated with lower speech intelligibility (Kim et al., 2011; Turner et al., 1995). It should be noted that the relatively high SI scores observed in the G6070 group may reflect the combined influence of the clinical recruitment setting, the demanding

nature of the connected speech task, and rater stringency, rather than indicating pathological speech deterioration. Future studies should include community-dwelling older adults and incorporate multiple SI assessment methods, such as transcription-based measures and visual analogue scales, to enable more precise interpretation of age-related changes in speech intelligibility.

In contrast, the F2 ratio showed no significant age-related differences in our cohort. This is notable, as the F2 ratio has proven effective in distinguishing healthy from pathological speech in other populations (e.g., Parkinson’s Disease, Down’s syndrome) (Corrales-Astorgano et al., 2018; Sapir et al., 2007). Our negative finding suggests that while the F2 ratio is sensitive to overt pathology, it may be less sensitive to the more subtle articulatory changes associated with healthy aging.

Finally, while this study confirms age-related acoustic changes, the specific variables predicting SI are known to be language-specific. For instance, one study found that while acoustic vowel space is a primary predictor in American English, Korean SI prediction also requires metrics like voice onset time and articulation rate (Kim & Choi, 2017). This linguistic dependency is a critical consideration for clinical applications and future research aiming to refine SI prediction models.

This study confirmed significant sex differences across all acoustic variables (formants, VSA, VAI, FCR, F2 ratios) and SI scores. Specifically, higher formant frequencies in females and a more posterior VSA distribution in males were observed, the latter being consistent with prior Korean data (Yoon & Kim, 2015). These acoustic differences are largely attributed to known physiological variations, such as vocal tract length (Jacewicz et al., 2007). This anatomical basis extends to perceptual findings, as women also demonstrated significantly higher SI scores, aligning with previous research (Kwon, 2010). A key finding was the significant correlation between acoustic metrics and perceived intelligibility, with VSA_5 showing the highest correlation with SI ($r = -.56$). However, it should be acknowledged that VSA_5 accounts for approximately 31% of the variance in perceived intelligibility ($r^2 = .31$), and the remaining variance may be attributable to other factors such as speech rate, voice quality, and prosodic control. These findings nonetheless suggest that VSA_5 holds promise as a meaningful complementary objective measure of SI in Korean speakers. The

superior predictive validity of VSA_5 may be attributed to the characteristics of the Korean vowel system. Unlike triangle- or quadrilateral-based metrics anchored primarily by high front and low vowels, VSA_5 incorporates all five monophthongs including the back vowels /o/ and /u/, thereby capturing both anterior-posterior and high-low articulatory dimensions more comprehensively. Given the relatively dense back and central vowel inventory in Korean, any articulatory deterioration across the full vowel space is more sensitively reflected in VSA_5, which likely accounts for its stronger correlation with perceived speech intelligibility. This highlights the potential of VSA_5 as an objective predictor of SI in Korean speakers.

From a clinical standpoint, VSA_5 holds particular promise as a practical tool in speech-language pathology. In the assessment of dysarthric patients, where perceptual SI ratings can be time-consuming and subject to listener variability, VSA_5 offers a rapid, objective, and non-invasive alternative that can complement subjective evaluation. Furthermore, its sensitivity to articulatory precision makes it well-suited for monitoring treatment outcomes in voice and speech therapy; SLPs can track changes in VSA_5 across sessions to quantify articulatory improvement and provide patients with tangible evidence of progress. In the context of voice training programs, SLPs may establish VSA_5-based benchmarks as measurable therapy goals, thereby enhancing the objectivity and accountability of rehabilitation planning. Given that VSA_5 incorporates all five Korean monophthongs, it may also capture a broader range of articulatory movement compared to triangle- or quadrilateral-based metrics, further supporting its utility as a comprehensive index of speech motor function. The utility of acoustic measures as predictors is well-established, and our findings reinforce the benefit of integrating objective acoustic analysis (like VSA) with subjective SI assessments to enhance clinical speech evaluation.

This study makes several notable contributions to the field. First, it provides large-scale normative acoustic data for native Korean speakers across age groups and sex, addressing a gap in the literature where most existing normative studies have focused predominantly on English-speaking populations. Second, by systematically comparing three VSA configurations (VSA_3, VSA_4, and VSA_5), this study empirically demonstrates that

VSA_5, which incorporates all five Korean monophthongs, yields the strongest correlation with perceived speech intelligibility, thereby proposing an optimized VSA metric tailored to the structural characteristics of the Korean vowel system. Third, the large-scale clinical dataset collected in an otolaryngology setting enables the establishment of age- and sex-stratified reference values that can be directly applied in clinical practice for the differential diagnosis of speech and voice disorders and for monitoring rehabilitation outcomes. Collectively, these findings advance our understanding of age- and sex-related acoustic variation in Korean and offer a practical foundation for objective speech assessment in both research and clinical contexts.

This study has several limitations. First, the exclusion of participants under 20 years of age prevents a comprehensive lifespan analysis; future research should incorporate childhood and adolescent data. Second, while a Likert scale was used for SI assessment, simultaneously employing additional measures such as orthographic transcription, phonetic transcription, or Visual Analogue Scale (VAS) could provide more comprehensive clinical insights. Third, participants were guided solely by a visual orthographic cue (‘ㅇ’) for the elicitation of /e/, without explicit differentiation from /ae/. Given the well-documented merger of /e/ and /ae/ in contemporary Korean, particularly among younger speakers, some acoustic overlap between these two vowels cannot be ruled out, which may have introduced variability in F1 values in the G2030 group. Future studies should incorporate protocols that explicitly elicit and perceptually verify /e/ and /ae/ as distinct vowels. Fourth, as participants were predominantly recruited from a hospital in the Gyeongsang dialect region, the normative values presented in this study may reflect regional dialectal characteristics rather than Standard Korean. Future research should include speakers from diverse dialectal backgrounds to enhance the generalizability of the findings.

Despite these limitations, this study provides robust, large-scale normative data confirming that age and sex significantly impact Korean vowel acoustics. Crucially, the finding that VSA_5 shows a meaningful correlation with subjective intelligibility and serves as a promising complementary objective measure offers a valuable resource for clinical and pedagogical application. Clinically, this non-invasive metric can be utilized for objective assessment, dif-

ferential diagnosis of speech and voice disorders, and monitoring rehabilitation progress. Pedagogically, SLPs can employ VSA_5 to establish measurable goals and track improvement in voice training programs.

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Appendix 1. Praat script for formant analysis

```
# Measure Formant for female
soundName$ = selected$ ("Sound")
Rename... soundObj
select Sound soundObj

To Formant (burg): 0, 5, 5500, 0.025, 50
f1 = Get mean: 1, 0, 0, "hertz"
f2 = Get mean: 2, 0, 0, "hertz"

pause Formant analysis complete. Proceed to the next step?

clearinfo
printline F1 is 'f1:2' Hz
printline F2 is 'f2:3' Hz

##### End of Script #####
```

국문초록

한국어 모음의 표준 포먼트 데이터: 임상 및 교육적 적용을 위한 자료

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배경 및 목적: 본 연구는 건강한 한국 성인의 모음 포먼트 주파수와 말 명료도에 대한 대규모 표준 데이터를 확립하여, 언어치료 및 음성과학 분야의 임상 및 교육적 적용을 위한 중요한 자료를 제공하는 것을 목표로 한다. **방법:** 세 개의 구별되는 연령 집단에 걸쳐 4,235 명의 건강한 한국 성인을 대상으로 포괄적인 횡단면 분석을 수행하였다. 맞춤형 Praat 스크립트를 사용하여 F1, F2, 모음 공간 면적 (VSA_3, VSA_4, VSA_5), 모음 조음 지수, 포먼트 중앙화 비율, F2 비율을 포함한 여러 음향 변수를 체계적으로 측정하였으며, 신뢰할 수 있는 기준선 데이터를 확립하기 위해 지각적 측정을 통해 주관적 말 명료도를 평가하였다. **결과:** F2 비율($p = .14$)을 제외한 모든 변수에서 유의미한 연령 관련 차이가 발견되었다. /o/의 F1 ($p = .327$)을 제외한 모든 변수에서 유의미한 성별 기반 차이가 확인되었다. 모든 음향 지표는 말 명료도와 유의미한 상관관계를 보였으며, VSA_5가 모든 모음 공간 지표 중 가장 강력한 예측력을 보여($p < .01$), 평가 목적을 위한 우수한 임상적 유용성을 나타냈다. **논의 및 결론:** 본 연구는 연령과 성별이 한국어 모음 음향학에 중요하게 영향을 미친다는 것을 확인하는 견고한 대규모 표준 데이터를 제공하며, VSA와 말 명료도의 연령 관련 감소를 밝혔다. 결정적으로, 이러한 발견들은 VSA_5를 주관적 명료도의 강력하고 객관적인 예측 변수로 확립하여, 임상의를에게 음성 평가, 감별 진단, 그리고 언어 및 음성 장애를 가진 다양한 임상 집단의 재활 진행 상황 모니터링을 위한 가치 있고 비침습적인 지표를 제공한다.

핵심어: 포먼트 주파수, 모음, 말 명료도

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